ATLS

Primary Survey

X - eX**sanguination**

A - Airway

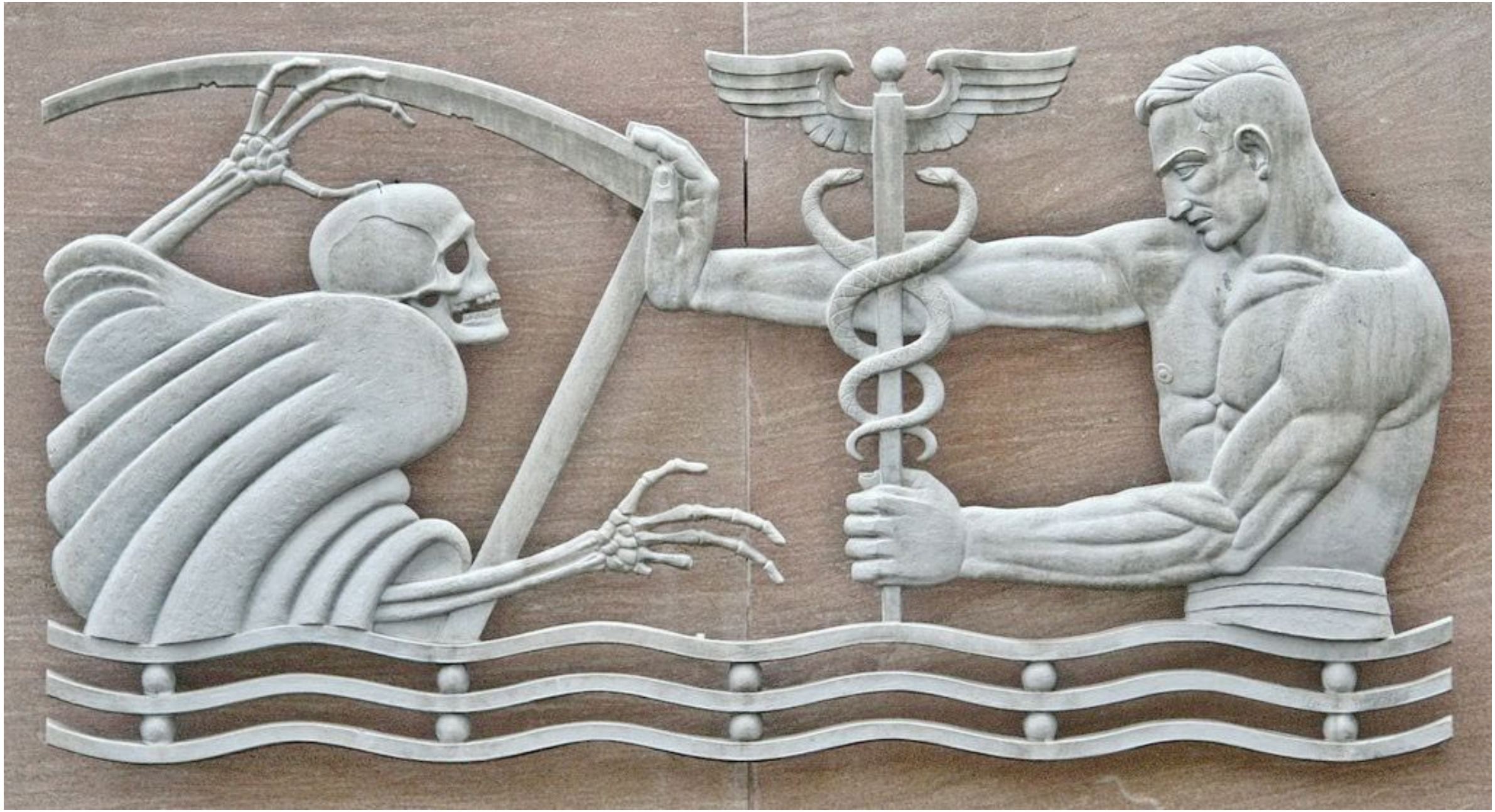
B - Breathing

C - Circulation

D - Disability

E – Expose/environment

brembo
5
HUBLOT
Ray-Ban
PIRELLI



Immediately Life-threatening Injuries

X - Massive external hemorrhage

A - Airway obstruction

B - Tension pneumothorax

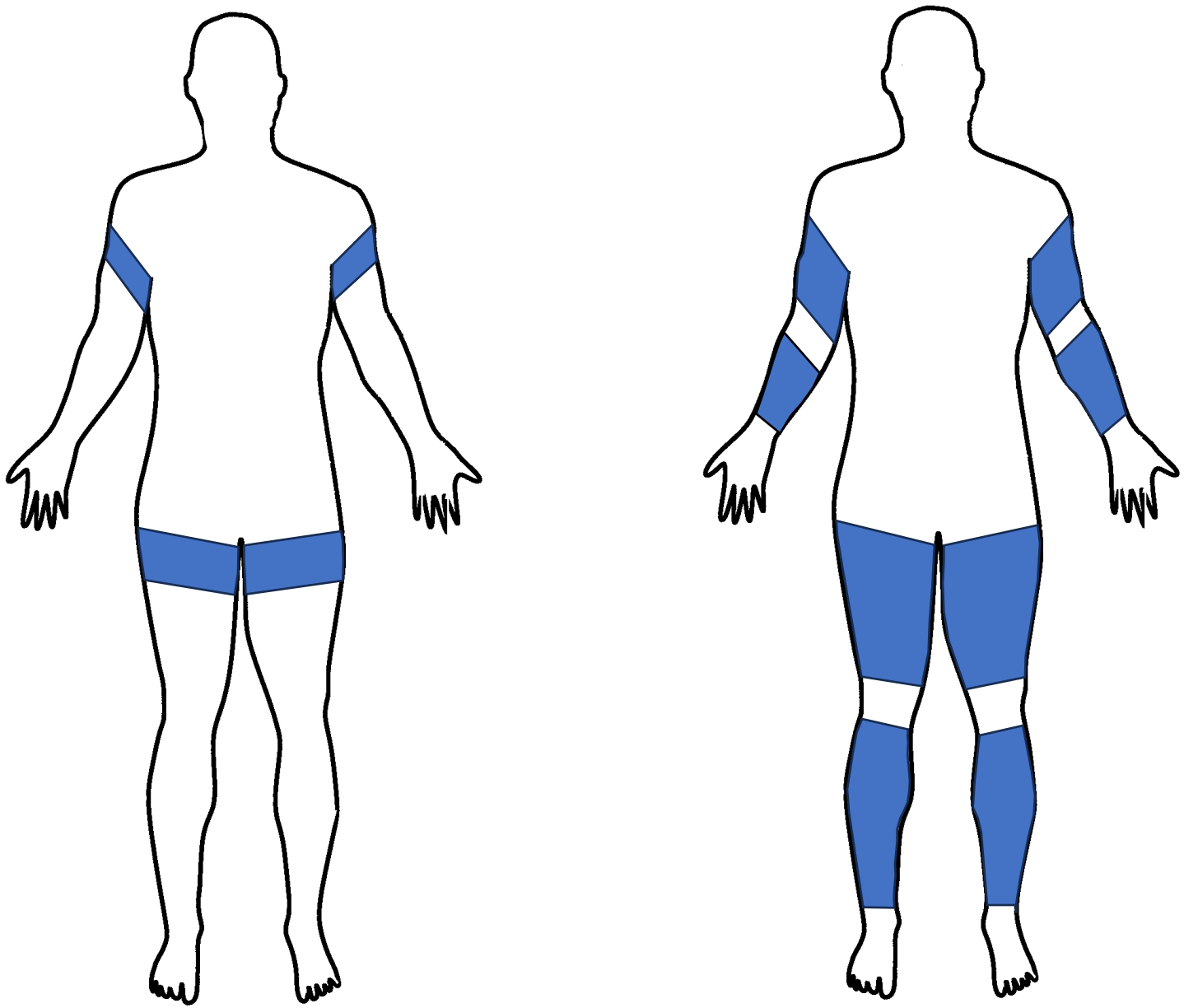
C - Cardiac tamponade

C – Hemorrhagic shock / Internal bleeding

D - Closed head injury

E - Neurogenic shock

eXsanguinating hemorrhage



TCCC Guidelines

DeployedMedicine.com

Limb tourniquets and junctional tourniquets should be converted to hemostatic or pressure dressings as soon as possible ***if three criteria are met:***

1. *the casualty is not in shock;*
2. *it is possible to monitor the wound closely for bleeding; and*
3. *the tourniquet is not being used to control bleeding from an amputated extremity.*

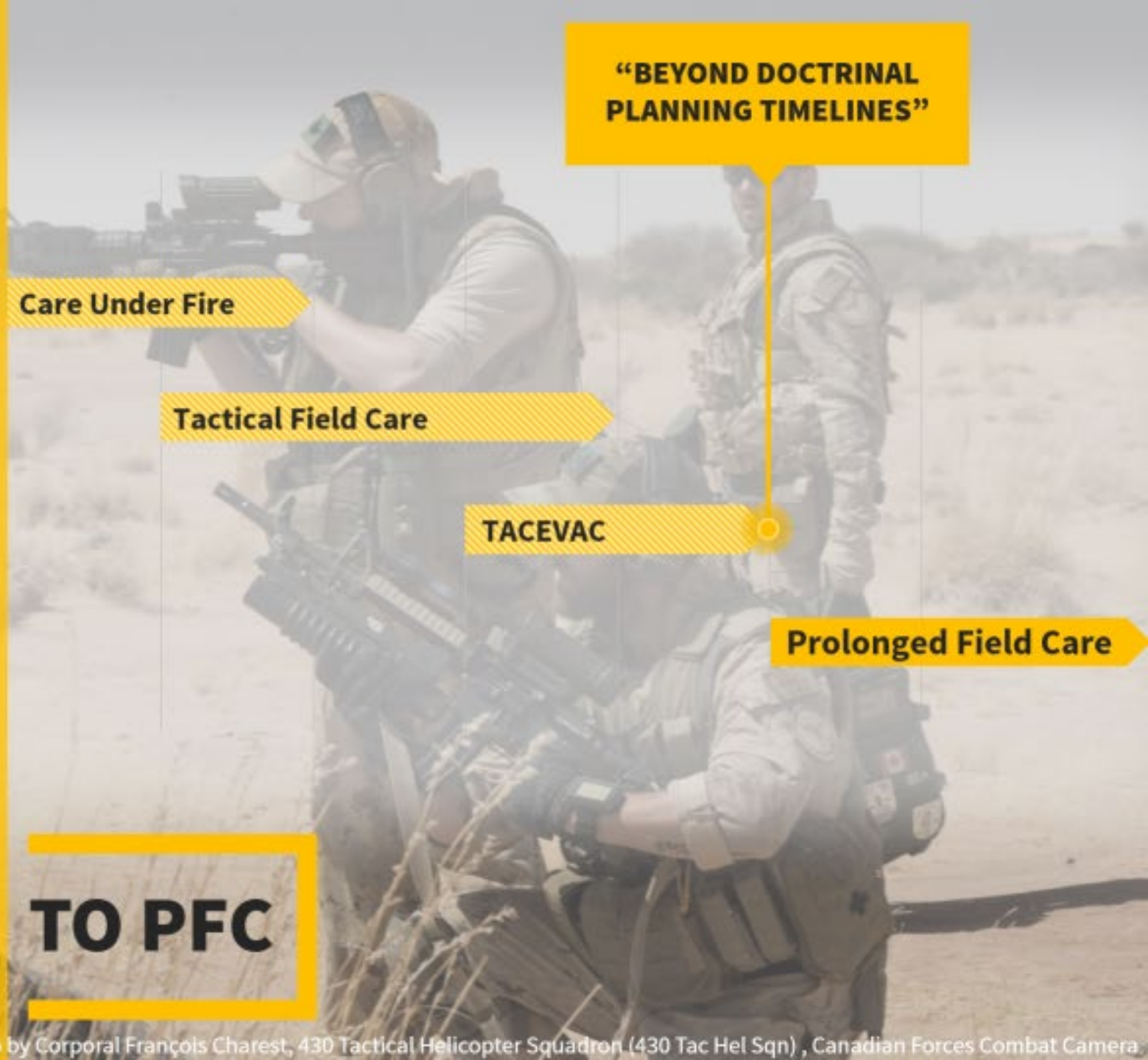
Every effort should be made to convert tourniquets in less than 2 hours if bleeding can be controlled with other means.

Do not remove a tourniquet that has been in place more than 6 hours unless close monitoring and lab capability are available.

PROLONGED FIELD CARE

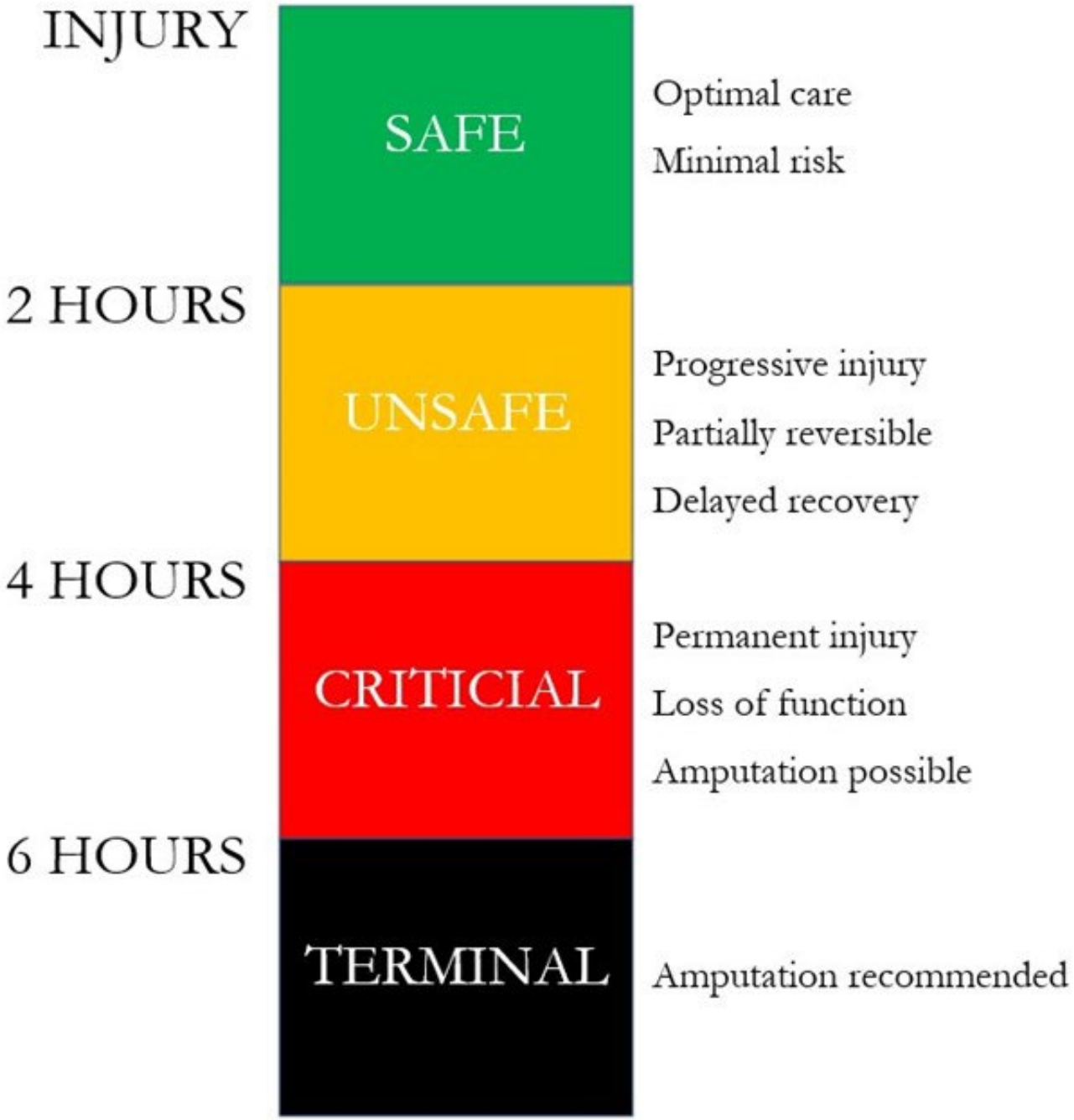
"Holding onto a **sicker** patient than you can care for, for **longer** than you want, with **fewer resources** than you need, in a **place** you don't want to be"

"You do not PLAN to have PFC. If your plan includes PFC, make a new plan"



TCCC TO PFC

Photo by Corporal François Charest, 430 Tactical Helicopter Squadron (430 Tac Hel Sqn) , Canadian Forces Combat Camera



Weinrauch 2023

Conversion Success

Lakstein et al 2003

IDF 4-year review

- 550 patients treated prehospital by IDF providers
- 16% of prehospital injuries had TQ applied
- 76% of prehospital TQ conversions successful

Ukraine FST Review

- Review by vascular surgeon of all TQ applied at single FST
- 25% of TQ applications “appropriate”

Yatsun 2024

Tissue Ischemia

Reperfusion Syndrome

Limb Salvage

Reperfusion Injury

- Metabolic Acidosis
- Cardiac arrythmia
- Hypotension
- Hypothermia
- AKI

Resuscitation Toolbox (AUZ Army)

Weinrauch and Peters 2023

A
- ALKALOSIS
- Hyperventilate. Sodium bicarb 8.4% 1 x 50ml ampule 5-10 minutes prior to release

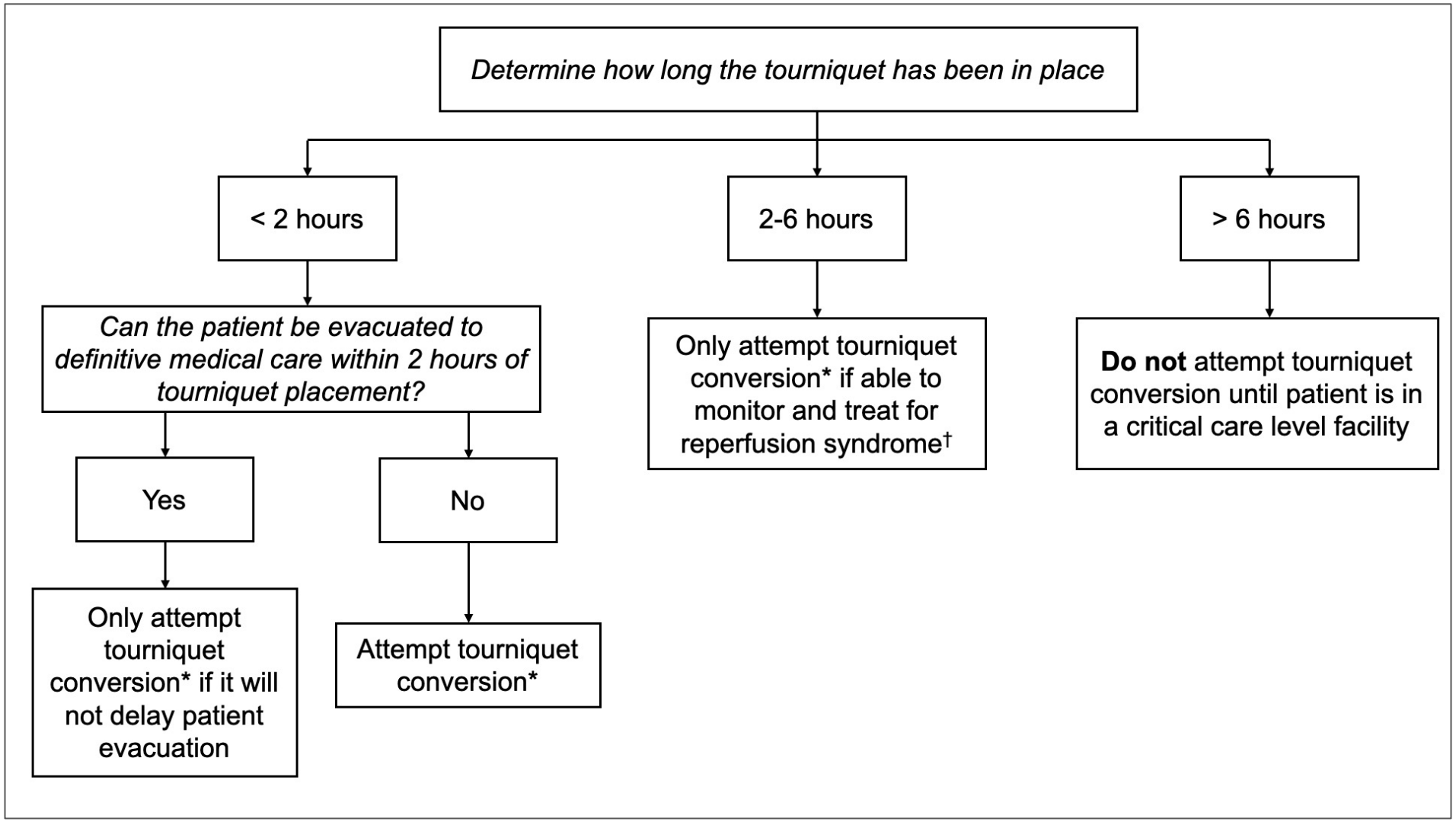
B
- BLOOD PRESSURE
- High enough to anticipate reperfusion drop

C
- CALCIUM
- Calcium Gluconate 10% solution 10ml over 2-3 minutes prior to release

D
- DEFIBRILLATION
- Pads applied prior to TQ release

E
- ECG/ ELECTROLYTES/ ENVIRONMENT
- Serum potassium low range (glucose/ insulin infusion)/ Warming devices

Permanent Damage – minimal Reperfusion injury – minimal	**<2 Hours**	• Attempt conversion • No need to manage reperfusion syndrome
Permanent Damage – likely Reperfusion injury – possible	**2-6 Hours**	• Attempt conversion in facility capable of ACLS care • **Be prepared to** treat reperfusion syndrome
Permanent Damage – likely Reperfusion injury – likely	**>6 Hours**	• Only attempt conversion in facility capable of ACLS care and surgery • **Pre-treat** for reperfusion syndrome



Jerome 2025

*Tourniquet conversion should only be attempted if no contraindications are present. The contraindications are the patient is in shock, the tourniquet is managing a traumatic amputation, or the clinician is unable to closely monitor the patient for ongoing bleeding after tourniquet removal.

† The necessary resources include cardiac monitoring, the ability to administer intravenous or intraosseous fluids, and the availability of advanced cardiac life support medications.

Immediately Life-threatening Injuries

X - Massive external hemorrhage

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C – Hemorrhagic shock / Internal bleeding

D - Closed head injury

E - Neurogenic shock

Airway

Breathing

Smaller bore chest tube (20-22 Fr) appropriate for trauma

- Non-inferior to large bore (28 Fr)
- Better tolerated

All patients receiving a surgical chest tube should receive 1 dose of antibiotic to prevent empyema (EAST, 2022)

- Ceftriaxone 2g IV x1

Circulation

TXA 2g slow IV push

- Compared to 1g IV push and 1g infusio
  - Same mortality
  - Same VTE rates

TXA IM injection

- Not recommended yet…

Circulation

Shock Index

- HR/SBP

Shock Index > 1

- HR > SBP
- More sensitive than ABC score for predicting need for MTP

Disability

GCS 13: E3, V5, M5

INTERNATIONAL STANDARDS FOR NEUROLOGICAL CLASSIFICATION OF SPINAL CORD INJURY (ISNCSCI)

ASIA — American Spinal Injury Association

ISCOS — International Spinal Cord Society

Patient Name _____ Date/Time of Exam _____

Examiner Name _____ Signature _____

RIGHT

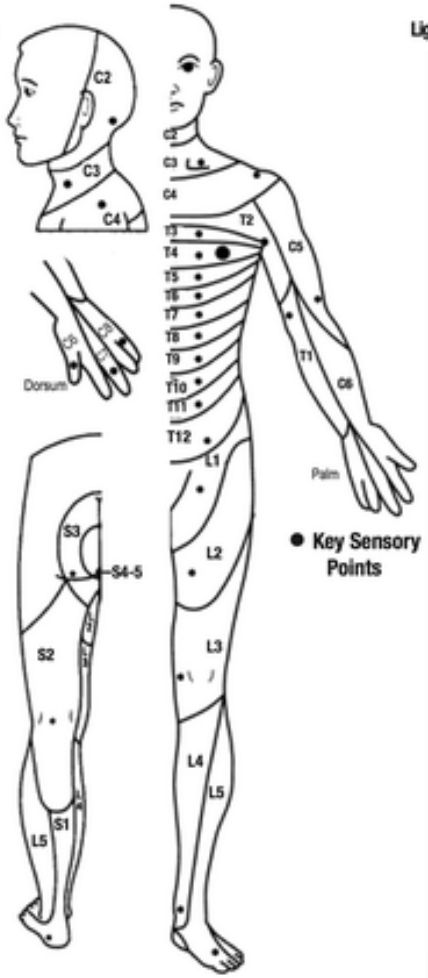
MOTOR KEY MUSCLES		SENSORY KEY SENSORY POINTS Light Touch (LTR)	Pin Prick (PPR)
	C2		
	C3		
	C4		
Elbow flexors	C5		
Wrist extensors	C6		
Elbow extensors	C7		
Finger flexors	C8		
Finger abductors (little finger)	T1		
	T2		
	T3		
	T4		
	T5		
	T6		
	T7		
	T8		
	T9		
	T10		
	T11		
	T12		
	L1		
Hip flexors	L2		
Knee extensors	L3		
Ankle dorsiflexors	L4		
Long toe extensors	L5		
Ankle plantar flexors	S1		
	S2		
	S3		
	S4-5		
RIGHT TOTALS (MAXIMUM)		(50)	(56)	(56)

UER (Upper Extremity Right)

LER (Lower Extremity Right)

Comments (Non-key Muscle? Reason for NT? Pain?):

(VAC) Voluntary Anal Contraction (Yes/No) ☐



LEFT

SENSORY KEY SENSORY POINTS Light Touch (LTL)	Pin Prick (PPL)	MOTOR KEY MUSCLES	
		C2	
		C3	
		C4	
		C5	Elbow flexors
		C6	Wrist extensors
		C7	Elbow extensors
		C8	Finger flexors
		T1	Finger abductors (little finger)
		T2	
		T3	
		T4	
		T5	
		T6	
		T7	
		T8	
		T9	
		T10	
		T11	
		T12	
		L1	
		L2	Hip flexors
		L3	Knee extensors
		L4	Ankle dorsiflexors
		L5	Long toe extensors
		S1	Ankle plantar flexors
		S2	
		S3	
		S4-5	
(56)	(56)	(50)	LEFT TOTALS (MAXIMUM)

UEL (Upper Extremity Left)

LEL (Lower Extremity Left)

MOTOR (SCORING ON REVERSE SIDE)

0 = total paralysis
1 = palpable or visible contraction
2 = active movement, gravity eliminated
3 = active movement, against gravity
4 = active movement, against some resistance
5 = active movement, against full resistance
5* = normal corrected for pain/disuse
NT = not testable

SENSORY (SCORING ON REVERSE SIDE)

0 = absent | 2 = normal
1 = altered | NT = not testable

☐ (DAP) Deep Anal Pressure (Yes/No)

MOTOR SUBSCORES

UER ☐ + UEL ☐ = UEMS TOTAL ☐ (MAX (25) (25) (50))

LER ☐ + LEL ☐ = LEMS TOTAL ☐ (MAX (25) (25) (50))

SENSORY SUBSCORES

LTR ☐ + LTL ☐ = LT TOTAL ☐ (MAX (56) (56) (112))

PPR ☐ + PPL ☐ = PP TOTAL ☐ (MAX (56) (56) (112))

NEUROLOGICAL LEVELS
Steps 1-5 for classification as on reverse

	R	L
1. SENSORY		
2. MOTOR		

3. NEUROLOGICAL LEVEL OF INJURY (NLI) ☐

4. COMPLETE OR INCOMPLETE? ☐
Incomplete = Any sensory or motor function in S4-5

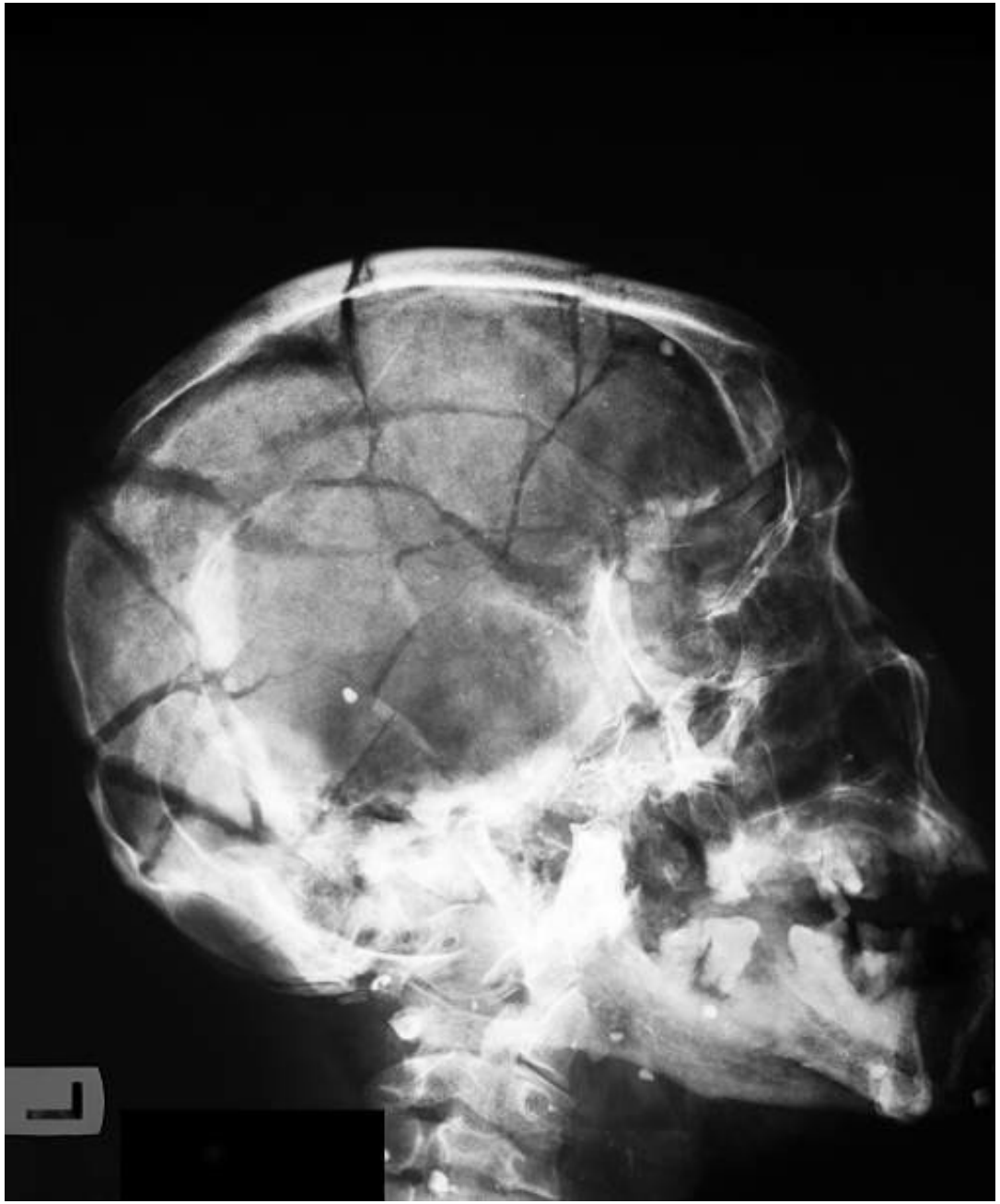
5. ASIA IMPAIRMENT SCALE (AIS) ☐

(In complete injuries only) **ZONE OF PARTIAL PRESERVATION**
Most caudal level with any innervation

	R	L
SENSORY		
MOTOR		

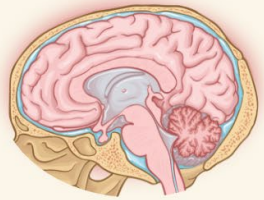
This form may be copied freely but should not be altered without permission from the American Spinal Injury Association.

REV 11/15

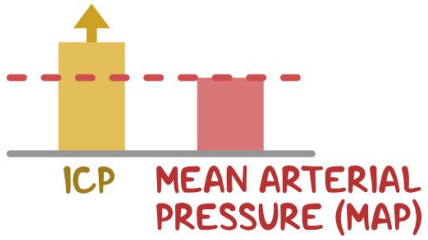


↑ INTRACRANIAL PRESSURE (ICP)

⚠ MEDICAL EMERGENCY ⚠



CAN RESULT from:
HEAD INJURY, BLEEDING in BRAIN, TUMORS, INFECTIONS, EXTRA FLUID in BRAIN, STROKE



WHEN ICP > MAP, BRAIN CAN NO LONGER RECEIVE ENOUGH O_2

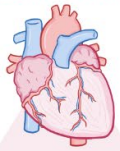
↳ SYMPATHETIC NERVOUS SYSTEM ACTIVATES

↳ THEN PARASYMPATHETIC NERVOUS SYSTEM ACTIVATES

RESULTS in →

CUSHING'S TRIAD

↓ HEART RATE



IRREGULAR RESPIRATION

WIDENED PULSE PRESSURE
(↑ difference between systolic & diastolic blood pressure)

OTHER SYMPTOMS of ↑ ICP

WEAKNESS, LETHARGY, HEADACHES, VOMITING, BLURRED VISION, CHANGES in BEHAVIOR

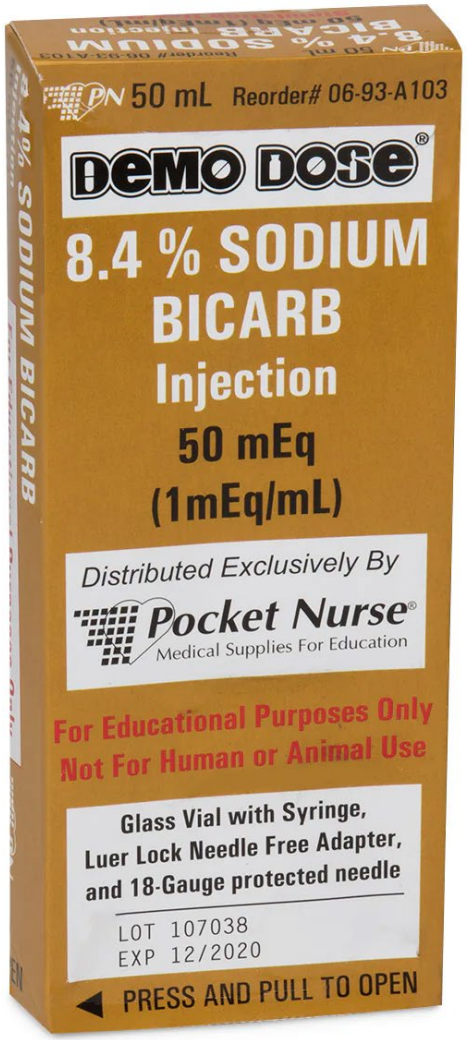
TREATMENT

AIMED at ↓ ICP, REVERSING CAUSE of ↑ ICP, and RESTORING BLOOD FLOW

* INTRAVENOUS MANNITOL
* HYPERVENTILATION
* ELEVATION of HEAD
* ADDITIONAL MEDICATION
* SOMETIMES: DRAINING EXTRA FLUID
* RARELY: CRANIOTOMY

OSMOSIS.org

100cc NaCl 3% = 1 amp biarb



5 Ways to Decrease Intracranial Pressure Using the Monro-Kellie Doctrine

- Enhance venous drainage
  - Elevate head 30°
  - If in a cervical collar, check fit
- Hyperosmolar therapy
- Hyperventilation

Environmental/Expose

Environment is important

Much easier to prevent heat loss than treat hypothermia

MTF in the upcoming hypothermia lecture

Expose and keep warm

Transition to secondary survey

Summary

ABC MOVIES

ACLS

- Full Code Pro app
- Vector change
- SVT vagal maneuvers and adenosine tricks

ATLS

- TQ application and conversion
- Antibiotics after every surgical chest tube
- Bicarb as hypertonic saline substitute

ANY
QUESTIONS?